

Automated Scoring of Asynchronous Interview Videos Based on Multi-Modal Window-Consistency Fusion

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Abstract—Soft skills, such as personality characteristics, communication skills and leadership, affect personal career performance greatly. Therefore, predicting the soft skills of interviewees can provide interviewers with a strong reference for the decision of hiring. Nowadays, as asynchronous video interviews have gradually become a popular form of interviews, automatic interview evaluation of soft skills has attracted widespread attention from researchers. However, existing automatic evaluation methods have two significant drawbacks. First, most of them model the problem as multi-modal fusion of long-term sequences, while ignoring the consistency of multi-modal expression in short-time windows, which is a key attribute of the interview scene. Second, without embedding of professional knowledge in the interview field, the interpretability of the model is relatively weak. To address the above problems, we propose a novel *Multi-modal Window-Consistency Fusion* network, namely MWCF, to capture the expression consistency of different modalities in a short-time window and re-weight the language signals to enhance important portions in verbal clues. Meanwhile, in order to enhance the interpretability of the evaluation model, we introduce the professional knowledge of interviewers by proposing a topic generation module based on question attention, and embedding the most representative keywords under different soft skills into the model. Furthermore, a real-world interview dataset is built by developing an asynchronous interview platform, and extensive experiments are conducted to show the superior performance of our proposed model.

Index Terms—Keyword attention, multi-modal interaction, natural language processing, social computing, time window, user modeling.

I. INTRODUCTION

VIDEO-BASED interviews have been used by more and more companies in the recruitment process [1][2] [3]. Compared with offline interviews, this kind of interview methods are more flexible and user-friendly. Interviewees needn't go to the city where the applicant company is located and can participate in the remote interview in a low-cost manner [4].

Manuscript received 29 June 2022; revised 1 May 2023; accepted 4 July 2023. Date of publication 11 July 2023; date of current version 2 September 2024. This work was supported by the Science and Technology Program of Guangzhou, China under Grant 2023B03J1388. Recommended for acceptance by C. Busso. (Corresponding author: Jianming Lv)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TAFFC.2023.3294335>, provided by the authors.

Digital Object Identifier 10.1109/TAFFC.2023.3294335

Interviewers can better coordinate the interview time of different interviewees to improve the efficiency of recruitment [5]. Video-based interviews can be classified into two categories: synchronous video interviews (both interviewees and interviewers are online at the same time) and asynchronous video interviews (interviewers are not online with interviewees). Compared with synchronous video interviews, asynchronous video interviews are preferred by many companies as a preliminary screening tool through a structured interview format [6][7] [8] [9]. In this asynchronous manner, human resources consultants set a collection of questions on the recruitment system in advance, and then send emails or SMS to invite interviewees to log in to the recruitment system for interview recording [10]. Follow-up interviewers also log in to the system to conveniently evaluate the interviewees and decide who can enter the next round of interviews.

The interviewee's soft skills such as adaptability, social expression, and leadership are important parts of evaluation criteria [11]. Interviewers make decisions based on interviewees' verbal and non-verbal behaviors in the recruitment process [12]. However, it is very challenging to establish the relationship between the soft skills and the verbal cues or non-verbal cues from social interaction [13][14] [15]. Therefore, the use of multi-modal models to automatically assess interviewees' soft skills is an interesting research topic in the field of social computing. This technology can not only provide interviewers with a fine-grained assessment of interviewees' ability to assist hiring decision [16][17] [18], but also help interviewees to improve their expressive skills through interview practice.

Nowadays, the development of automatic evaluation models still has the following limitations. 1) *Lack of interpretability*: Most of the existing automatic evaluation models have low interpretability and can not combine with prior professional knowledge of human resources consultants. 2) *Ignoring short-time consistency of multiple modalities*: Most existing research focuses on solving the multi-modal fusion problem of long-term sequences, while ignoring the important short-time consistency between different modalities of an interviewee's expression. For example, when an interviewee says that he is a confident and cheerful person, his voice becomes trembling and quiet. This shows the inconsistency between the speech content and the behavior shown, thus revealing the contradiction of what the interviewee said at that moment.

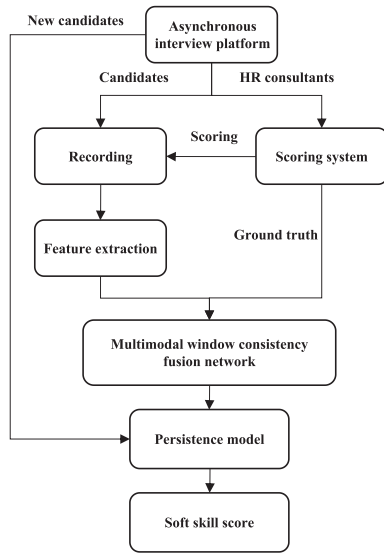


Fig. 1. The overview of the task framework for automatic interview evaluation.

In this paper, a *Multi-modal Window-Consistency Fusion* (MWCF) model is proposed to solve the above challenges. In particular, a topic generation module (TGM) based on question attention is proposed to improve the interpretability of the model, which introduces HR consultants' professional experience and embeds the most representative keywords under different soft skills. Furthermore, the window-consistency of multi-modal information is proposed to consider the short-time consistency between different modalities, and re-weight the language signals to enhance important portions.

The procedure to optimize and deploy the MWCF model is shown in Fig. 1. First of all, we cooperate with human resource companies (HR) to build an asynchronous online interview platform for structured interviews in real scenes, where candidates are invited to log in to the system to record their answers to some preset questions. Then HR consultants score the soft skills of interviewees according to some hiring standards. The scored records serve as labeled data to train the MWCF model, which also integrates with the professional prior knowledge from HR departments. Finally, for new coming interviewees, the optimized model is applied to predict the soft-skill scores automatically.

The major contributions of this paper are summarized as follows:

- A multi-modal scoring framework is proposed to evaluate interviewees automatically, which utilizes both verbal and non-verbal clues effectively to make a more precise decision than traditional single-modal systems.
- A novel *Multi-modal Window-Consistency Fusion* (MWCF) model is proposed to capture the short-time expression consistency of different modalities of interviewees' information, and re-weight the important portions in verbal clues with higher consistency to reduce the side-effects of noise.

- An attention-based topic generation model is proposed to integrate with the prior knowledge of HR consultants to achieve better interpretability of the evaluation model, which is able to show easily-verified keyword-based scoring proof. Extensive experiments show that the proposed method can effectively evaluate the different soft skill scores of interviewees with performance significantly better than the baseline methods.

II. RELATED WORK

A. Automatic Interview Evaluation

Most of the automatic interview evaluation methods [19][20] [21][22] are based on the structured interview, which adopts a set of preset questions and adopts the same test process to all applicants. Based on interviewees' answers to standardized questions, interviewer will score the knowledge, ability, quality and other aspects of the candidates. Compared with unstructured interviews, structured interviews are more fair and reliable, because each interviewer has a unified standard of evaluation, which avoids arbitrary questions without objective basis.

The core tasks of automatic interview evaluation include predicting the personality, fluency, or employability of interviewees. Most of the traditional automatic evaluation algorithms rely on cumbersome feature engineering, and utilize some manually defined rules to make predictions. In particular, Aydin et al. [23] used random forests to predict the personalities of characters in videos. Sinha et al. [24] collected corporate social media posts and used statistical features such as word similarity to analyze the six personality traits of employees (honest-Humility, emotion, extroversion, agreeableness, due diligence, and openness of experience). Nambiar et al. [25] used audio and vocabulary features, combined with linear regression, LDA (latent dirichlet allocation) topic generation model and Support Vectors Machines, to predict interviewees' fluency, self-confidence and other traits. Muralidhar et al. [26] collected 169 structured interview data sets through scenario simulation, and used Support Vector Machines to predict interviewees' enthusiasm.

More recently, deep neural networks based evaluation methods have been continuously developed. Specifically, Chen et al. [19] applied the neural network-based Doc2Vec model to predict interviewees' personality traits. Hemamou et al. [18] collected structured interview videos of the sales profession, and used the recurrent neural network to assess interviewees' employability. In this model, both language and non-verbal behaviors (facial expressions, speech) were combined to predict the employability of interviewees.

B. Attention Mechanism

Bahdanau et al. [27] introduced the attention mechanism for the first time in the field of natural language processing, and used it in the machine translation task based on the encoder-decoder framework. A lot of follow-up experiments have proved that the

attention mechanism can effectively screen out important information and improve the interpretability of neural networks [28]. Many researchers have proposed different attention calculation methods according to the specific characteristics of different NLP tasks. In particular, Lin et al. [29] proposed a self-attention mechanism to extract the interpretable embedding of sentences. Yang et al. [30] proposed a hierarchical attention mechanism to capture the hierarchical structure of words, sentences, and chapters in the document.

Inspired by the above research, the attention mechanism can be introduced in the neural network based automatic interview evaluation method to make the model pay more attention to the effective information delivered by interviewers. In particular, we propose a novel window-consistency attention mechanism in this paper, which captures the short-time consistency of multiple modalities and re-weights the important portions of verbal clues to improve the accuracy of final decision.

C. Multi-Modal Fusion

Nowadays, multi-modal information processing has become an essential task in most large-scale information systems thanks to the rapid development of the Internet of Things. Multi-modal fusion is a hot research topic in this field [31] [32] [33], which focuses on exploring the complementary relationship between various modalities to improve the performance of systems [34]. Classical multi-modal fusion methods can be divided into three categories roughly: feature-level fusion, decision-level fusion, and model-level fusion [35]. In particular, the feature-level fusion requires splicing, concatenating and re-weighting the features vectors of different modalities in the early stage [36]. The concatenation of multi-modal feature vectors in this kind of method can usually lead to redundant feature representation and make the model tend to overfit when the training set is small. The decision-level fusion achieves the final results by combining the outputs from sub-models of different modalities by weighted summation, averaging, or voting [37]. The limitation of this kind of method is ignoring the relationship between different modalities and treating them independently. The model-level fusion takes a more flexible combination of multiple modalities, which utilizes the information of one modality to guide the generation of another one [38] [39]. Wang et al. [40] proposed a multimodal fusion model for emotional analysis tasks, taking into account the fine-grained structure of nonverbal sequences, and dynamically changing the representation of words based on nonverbal cues to capture the dynamic characteristics of language. Hemamou et al. [41] propose a sequential and multimodal deep neural network model that combines word, voice, and facial expression information from interviewees' videos to predict the employability of the interviewees.

Most of the existing multi-modal automatic evaluation methods [16][17] [42] focused on the fusion of long-time series, while ignoring the short-time consistency of multi-modal information expressed by interviewees, which is also crucial clues to judge the soft skills of interviewees. Motivated by this observation, in this paper, we propose the window-consistency attention mechanism to extract higher-level semantic information by integrating it with multi-modal consistency information.

D. Ethics of AI in Human Resources

In addition to the automatic interview evaluation involved in this article, AI also permeates many other aspects of the human resources department, such as analyzing and screening the resume of job seekers [43], providing personalized career guidance [44], and predicting employee turnover [45]. There will inevitably be ethical problems. First of all, the algorithm may bring bias and discrimination. For example, the artificial bias in the data may be learned by the algorithm, such as age, gender and other aspects of discrimination, resulting in unfairness to some groups [46]. Second, in terms of data privacy, AI system needs to collect users' personal information, which may infringe personal privacy and data protection rights [47]. In addition, there is also the problem of transparency. Because AI decisions may use black-box algorithms, it is difficult for system developers to explain the decision-making process of the model, resulting in a lack of transparency [48].

In order to solve these problems, we can first establish a more fair data set to eliminate the possible discrimination problems, and regularly correct the bias [49]. Second, the AI system should comply with laws and regulations and take necessary measures to protect users' personal privacy [50]. For the interpretability of the model, AI should use more interpretive algorithms to make the decision more transparent [51]. Finally, the government and the industry also need to gradually improve the regulatory mechanism, and carry out a regulatory review on the AI interview system from resume screening, interviews and a series of links to ensure that it complies with relevant regulations and ethical standards [52].

In this paper, we propose an interpretable method that combines the professional prior knowledge of the human resources department. Through the attention mechanism based on soft skills keywords, we highlight keywords and video clips with high attention weight in our system. Interviewers can perform backtracking based on the highlighted segments, which improves the transparency of the algorithm to a certain extent. In addition, interviewers can regularly optimize data biases that may exist in the dataset from the system's scoring feedback to interviewees, and iteratively optimize the soft skill keyword knowledge base, thereby reducing the noise that may exist in the prior knowledge provided by human resources consultants at an early stage.

III. METHODOLOGY

A. Data Collection

Through cooperation with a human resources company, we have developed an online B/S platform to conduct asynchronous interviews. The purpose of this project is to model a total of 18 soft skills items, including abstract ability, achievement orientation, adaptability, analytical ability, command ability, conscientiousness, gregariousness, imaginativeness, independence, interpersonal sensitivity, methodicalness, perseverance, persuasive-ness, result-orientation, social confidence, social skills, thinking skills and trait competitiveness, for participants by using multimodal social signals in the video. In order to collect data, we conducted a mock interview, inviting interviewees with certain

work experiences in society who want to receive feedback and suggestions on soft skills items. The data collection process is as follows: First, the invited candidates log in to the interview system and record their responses to the questions in the system. They have one minute to think and five minutes to answer questions. The structured questions we set include the following questions.

- 1) Please introduce yourself in terms of personal background, work experience, career goals, and personal strengths.
- 2) Please describe the most successful thing you have done in your field. Tip: please use the STAR principle to describe the background situation, the specific tasks, the specific actions and the final results.
- 3) Describe a time when you encountered obstacles in persuading others to accept your idea. What was the situation then? Why didn't others agree with you? How did you deal with such obstacles? How did you influence other people's thoughts? What was the final result?
- 4) Describe a time when you had to establish a relationship with a stranger, who could be an intern, a member of a club, a mentor, etc. What was the situation then? How did you build a relationship with him / her? What methods did you use to promote the relationship? What was the final result?

Each candidate will be asked three questions. The first question (self-introduction) above is fixed, and then two randomly selected questions from questions 2-4 above will be asked. After the interview is completed, according to the soft skills scoring criteria, three human resources consultants will rate the candidate's overall performance and the performance of the above 18 soft skills items on each question (except for self-introduction), and we average the scores of all consultants as the final result. In addition, human resources consultants, through their professional experience, select the most representative keywords under different soft skills, and we integrate them into the model as professional prior knowledge for joint training. Please refer to the appendix for detailed soft skill scoring criteria and keywords, available online.

Finally, a total of 103 candidates' answers were collected. Due to the lack of camera permissions among some candidates, valid answers from 96 candidates were ultimately obtained as a fine-grained dataset after data cleaning. Before the implementation of the project, each participant in the experiment was informed that all data would be used during the implementation of the project, and we obtained the consent of each participant. Although candidates agree to use their interviews in the experiment, the collected dataset will not be released to the public due to privacy protection regulations for videos. After the project is completed, each participant will receive feedback from the interviewer and provide personalized suggestions to better improve their personal abilities.

B. Task Definition

Table I summarizes the notations used in methodology. In this paper, the automatic interview evaluation task is formally described as a regression task: given a structured interview

TABLE I
DESCRIPTION OF THE SYMBOLS USED IN THE MODEL

Symbol	Description
q_i	the i^{th} word of question sequence
m	length of question sequence
s_i	the i^{th} soft skill
$k_{i,j}$	the j^{th} keyword of the i^{th} soft skill.
b	length of keyword list
Y_i	interviewee's score of the i^{th} soft skill
c_i	topic vector of the i^{th} soft skill
M	dimension of word vector
B	dimension of visual vector
N	dimension of acoustic vector
n	length of modality sequence
e_t^x	the original feature vector at time step t , where $x \in \{V, A, W\}$ stands for different modalities
s_t^x	the feature vector after semantic alignment at time step t , where $x \in \{V, A, W\}$ stands for different modalities

question $Q = \{q_1, q_2, \dots, q_m\}$ composed of m words, the corresponding video of an interviewee's answer is denoted as V . The set of soft skills is denoted as $\Omega = \{s_1, s_2, \dots, s_l\}$, where $s_i (1 \leq i \leq l)$ means a specific soft skill, eg. adaptability, thoughtful, independence, etc. The keyword knowledge base of the soft skill s_i provided by the HR consultants is denoted as $K_i = \{k_{i,1}, k_{i,2}, \dots, k_{i,b}\}$, where b is the number of words in the keyword list. The goal of the model is to predict the corresponding interviewee's score Y_i of the soft skill s_i .

C. Data Preprocessing and Feature Extraction

From a candidate's video record, we can extract text-based features, acoustic-based features, and visual-based features for multi-modal analysis. The steps of data preprocessing and feature extraction are as follows.

Text-based Feature Extraction: Psychological studies have shown that language can reflect individual differences and similarities in thinking patterns [53], emotions [54] and social relationships [55]. For example, the people with social confidence tend to use more positive emotional language, and show more recognition and praise for others [56]. Therefore, we can extract the words from interviews aided by the automatic speech recognition technology [57] [58], and apply the pre-trained word embedding vectors [59] to achieve the text feature vectors, which are denoted as $\{e_t^W\}$. Here $t \in [1, n]$ indicates one time step of the input sequence, and is corresponding to one word in the speech content. This paper uses pre-trained 300-dimensional word vectors from Baidu Encyclopedia corpus.¹ The human resources company we worked with purchased a customized iFLYTEK ASR system² and deployed it on their server. This paper uses this service to transcribe the verbal content and extract the corresponding timestamp expressed by the interviewee. The system has an accuracy rate of over 97% for Mandarin recognition in a quiet environment. Therefore, before inviting interviewees, we require interviewees to answer questions in a quiet environment (such as at home or in a conference room) to avoid the impact of noisy environments on speech.

¹<https://github.com/Embedding/Chinese-Word-Vectors>

²iFLYTEK ASR system: <https://global.xfyun.cn/products/lfasr>

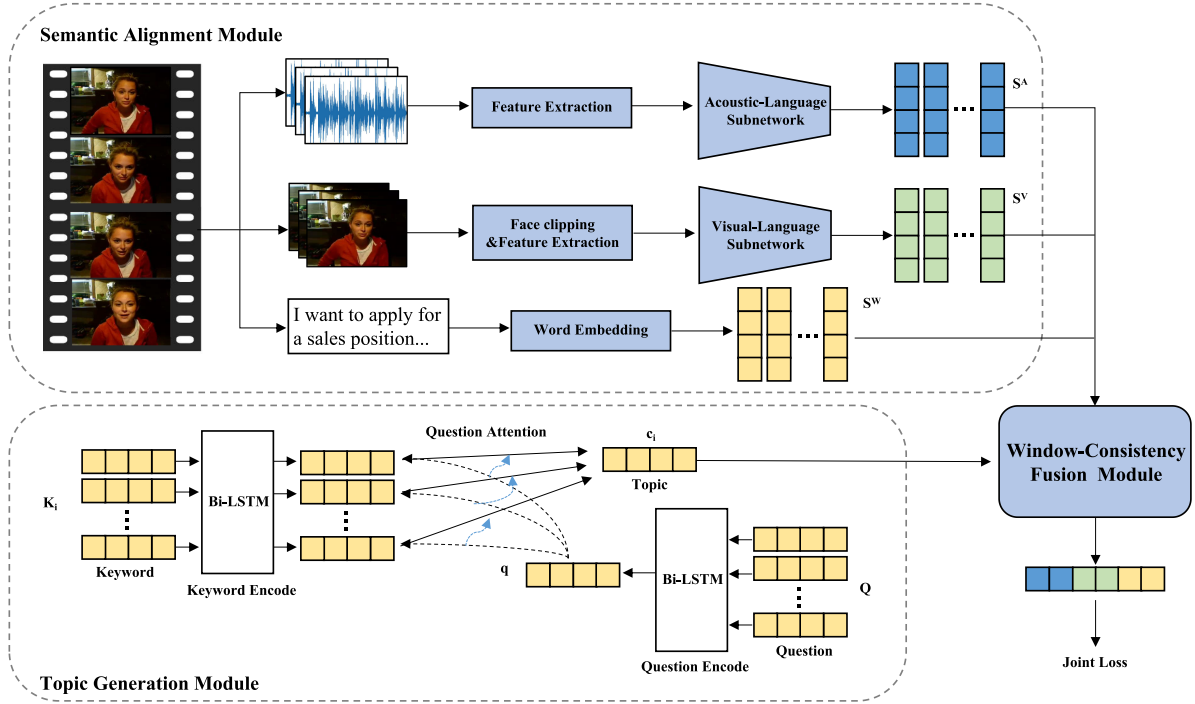


Fig. 2. Overview of the proposed method. According to the time of the word, we extract the corresponding video frame and audio frame in the asynchronous interview video. And in order to ensure that the three modals are in the same semantic space, we do nonlinear mapping for the extracted features. Finally, the topic vector c_i , visual features S^V , audio features S^A and word features S^W are input into the multi-modal window consistency fusion to output the calculation results.

Acoustic-based Feature Extraction: In addition to text features, acoustic features are also important cues in the interview scenes. Many researchers have proved that voice, speed, intensity and other voice signals play an important role in psychological behaviour analysis [60] [61] [62]. To align the acoustic signal with the text data, we separate the sound clips according to the word-level granularity in the text. For example, when the text "hello" said by the interviewee appears in 0.1-0.4 seconds of the video, the audio clip at 0.1-0.4 seconds of the video is extracted accordingly. We use the Opensmile Toolkit [63] to extract the acoustic low-level descriptors (LLDs), such as harmonics-to-noise ratio (HNR), zero-crossing-rate (ZCR) and mel-frequency cepstral coefficients (MFCC), comprising a total of 384 dimensions proposed in [64] as the segment features $\{e_t^A\}$, where $t \in [1, n]$ indicates one time step of the input sequence.

Visual-based Feature Extraction: Similar with the preprocessing steps of acoustic features, we first extract the image frames from the video according to the word-level granularity in the text, and select out one image randomly from the frames corresponding to each word in the speech content. The Dlib tool [65] is then used to clip the face region from the image frame for feature extraction. In order to capture the micro expression information of interviewees, we use the VGGNet [66] pre-trained on the CK+ [67] facial expression datasets, and extract the face features output by the last layer of the VGGNet as visual features, which are denoted as $\{e_t^V\}$. Here $t \in [1, n]$ indicates one time step of the input sequence.

D. Multi-Modal Window-Consistency Fusion Model

The proposed Multi-modal Window-Consistency Fusion Model, namely *MWCF*, is shown in Fig. 2, which includes the topic generation module, semantic alignment module and window-consistency fusion module. The details of these modules will be introduced in the following sections.

1) **Topic Generation Module:** The questions set in the interview are usually discussed around a certain topic. For example, when describing a scene of establishing a relationship with strangers, the topic discussed is related to interpersonal communication. In order to integrate the prior knowledge of the human resource department, we propose a topic generation module based on question attention. Specifically, the human resources department selects the most representative keywords for each soft skill through professional experience. The keywords of the soft skill s_i are denoted as $K_i = \{k_{i,1}, k_{i,2}, \dots, k_{i,b}\}$, where b is the number of words in the keyword list. The given structured question $Q = \{q_1, q_2, \dots, q_m\}$ is input into the word embedding layer and the following bi-direction long short term memory (Bi-LSTM) layer to achieve the encoded question representation q

$$q'_t = W_e q_t \quad (1)$$

$$\vec{h}_t = \overrightarrow{LSTM}(q'_t), t \in [1, m] \quad (2)$$

$$\overleftarrow{h}_t = \overleftarrow{LSTM}(q'_t), t \in [m, 1] \quad (3)$$

$$q = [\vec{h}_m, \overleftarrow{h}_m] \quad (4)$$

where q_t is denoted the one-hot representation of the word at the time step t of the question sequence. $W_e \in \mathbb{R}^{M \times d_{size}}$ is a pre-trained word embedding matrix, where M is the dimension of the word vectors and d_{size} is the length of the vocabulary.

Similarly, we also send the keywords of soft skills into the word embedding layer and Bi-LSTM to get the encoded representation of the j th keyword of the i^{th} soft skill, which is denoted as $u_{i,j}$

$$k'_{i,j} = W_e k_{i,j} \quad (5)$$

$$\overrightarrow{u_{i,j}} = \overrightarrow{LSTM}(k'_{i,j}), j \in [1, b] \quad (6)$$

$$\overleftarrow{u_{i,j}} = \overleftarrow{LSTM}(k'_{i,j}), j \in [b, 1] \quad (7)$$

$$u_{i,j} = [\overrightarrow{u_{i,j}}, \overleftarrow{u_{i,j}}] \quad (8)$$

where $k_{i,j}$ denotes the one-hot representation of the j^{th} keyword of the i^{th} soft skill.

Then we use the encoded question q as query, and the encoded keyword $u_{i,j}$ as key and value to calculate the question attention based topic vector c_i as follows:

$$\alpha_{i,j} = \frac{\exp(u_{i,j}^\top q)}{\sum_{t'=1}^b \exp(u_{i,t'}^\top q)} \quad (9)$$

$$c_i = \sum_{t'=1}^b \alpha_{i,t'} u_{i,t'} \quad (10)$$

2) *Semantic Alignment Module*: After extracting multi-modal features, in order to reduce the semantic gap between different modalities, we use two subnetworks namely *Visual-language Subnetwork* and *Acoustic-language Subnetwork* to project visual and acoustic features into the text feature space, and use them for subsequent multi-modal fusion. Specifically, we set the visual feature as $e_t^V \in \mathbb{R}^B$ and the acoustic feature $e_t^A \in \mathbb{R}^N$, where B, N represents the dimension of each modality and $t \in [1, n]$ is denoted as the time step of input sequence. The procedure of cross-modal projection is formulated as follows:

$$s_t^V = \phi(W_v e_t^V + b_v) \quad (11)$$

$$s_t^A = \phi(W_a e_t^A + b_a) \quad (12)$$

where $W_v \in \mathbb{R}^{M \times B}$, $W_a \in \mathbb{R}^{M \times N}$ are weight matrices, $b_v \in \mathbb{R}^M$, $b_a \in \mathbb{R}^M$ are weight vectors, and ϕ is the ReLU function. After above projection, different modalities are in the same subspace with the same dimension M .

3) *Window-Consistency Fusion Module*: In general, The potential personality characteristics and soft skills of interviewees can be more truly reflected in the moments with consistent representation of voice, speech content and face expression. In order to make use of the short-time consistency between multiple modalities, we propose the *Window-consistency Fusion Module* to strengthen the important portions with higher consistency. Specifically, denote $E = \{e_1^W, e_2^W, \dots, e_n^W\}$ as the sequence of words in the interviewee's speech content where n is the number of words. After a word embedding layer, the low-level

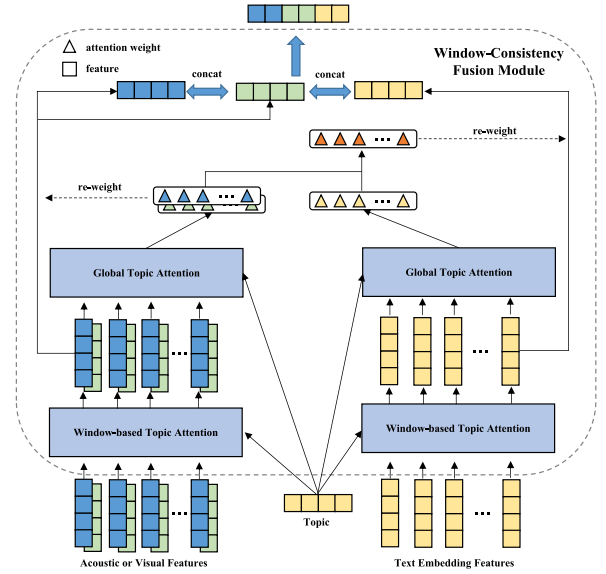


Fig. 3. Window-Consistency Fusion Module.

text feature vector can be obtained

$$s_t^W = W_e e_t^W \quad (13)$$

where W_e is a pre-trained word embedding matrix as the one used in (1). Then the text feature matrix $S^W = \{s_1^W, s_2^W, \dots, s_n^W\}$, the visual feature matrix $S^V = \{s_1^V, s_2^V, \dots, s_n^V\}$, and the acoustic feature matrix $S^A = \{s_1^A, s_2^A, \dots, s_n^A\}$ are fed into the window-consistency fusion module with the topic vectors c_i as shown in Fig. 2.

The detail of the *Window-consistency Fusion Module* is shown in Fig. 3. Specifically, each modality is fed into a window based topic attention layer to extract the localized representative features. The detail of the window based attention procedure is shown in Fig. 5, where a sliding window with the width w is applied on each single-modal sequence and the topic attention is calculated in each short-time window. Similar to the self-attention mechanism [68], the topic vector c_i (from (10)) is used as query Q , the single-modal feature matrix S^x (x is W, A or V) is used as value V and key K , and the window-based topic attention is calculated as follows:

$$A_{window}(Q, K) = \text{Softmax} \left(\frac{QK^T}{\sqrt{M}} \circ \text{Window-Mask} \right) \quad (14)$$

$$\text{WindowAtt}(Q, K, V) = A_{window}(Q, K) * V \quad (15)$$

$$H_i^x = \text{WindowAtt}(c_i^T \otimes n, S^x, S^x) \quad (16)$$

where H_i^x is the final single-modal representation related to the i^{th} soft skill, $x \in \{V, A, W\}$ indicates different modalities, M is the dimension of feature space, and $c_i^T \otimes n$ means producing a matrix by repeating the vector c_i^T for n times, $\circ$ means element-wise multiplication. The window mask here represents a local attention constraint as shown in Fig. 4, which illustrates an example of window mask with a window size of 1. The value of gray part is 1, the one of white part is 0, and the diagonal indicates

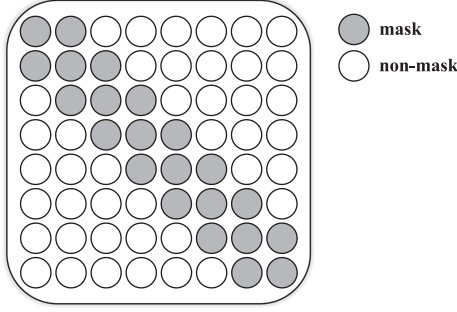


Fig. 4. Example of window mask with a window size of 1, the value of gray part is 1, and the value of white part is 0, diagonal indicates the current time step.

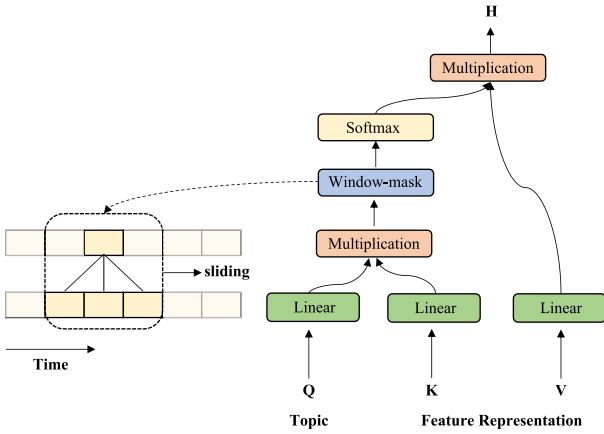


Fig. 5. Window-based topic attention.

the current time step. The use of window mask controls the size of the short-time window, within which the speech content usually focuses on the same topic.

In order to measure the consistency of the different modalities within each sliding window, we apply the global topic attention on H_i^x as shown in Fig. 6 to achieve the topic attention score on each sliding window. The sliding window with high score on all modality indicates a consistent expression, and requires more attention for making the final scores. Specifically, the topic attention score, which is denoted as A_{global}^x (x is A , V , or W indicating different modalities), is computed as follows:

$$A_{global}^x = \text{Softmax} \left(\frac{QK^T}{\sqrt{M}} \right) \quad (17)$$

where $Q = c_i^T$ and $K = H_i^x$. By combining the attention score of each modality, we can achieve the consistency score as

$$A_{cons} = (A_{global}^W + A_{global}^V + A_{global}^A) / 3 \quad (18)$$

A higher consistency score indicates the consistent attention on all modalities which may reflect the soft skill of the interviewee with higher possibility. Thus, we apply the consistency score to re-weight the sequence of speech content to achieve more reliable representation

$$T_i = A_{cons} * H_i^W \quad (19)$$

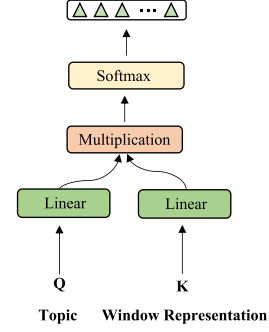


Fig. 6. Global topic attention.

Meanwhile, we also apply the attention score to re-weight the audio and visual modalities to achieve the final presentation G_i^x (x is A or V)

$$G_i^x = A_{global}^x * H_i^x \quad (20)$$

Finally, we concatenate the high-level semantic features of different modalities and input them into the multi-layer perceptron to predict the final score of a soft skill

$$Y_i = \phi(W_y[T_i; G_i^V; G_i^A] + b_y) \quad (21)$$

where W_y is weight matrices, b_y is bias, and ϕ is the ReLU function.

The joint loss function by combining all soft skills is formulated as the Huber Loss function [69]

$$Loss = \sum_{i=1}^l L(Y_i^{true}, Y_i) + \lambda \|\theta\| \quad (22)$$

$$L(y, Y) = \begin{cases} \frac{1}{2}(y - Y)^2, & \text{if } |y - Y| \leq \delta \\ \delta|y - Y| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases} \quad (23)$$

where l is the number of soft skills, Y_i^{true} stands for ground-truth, Y_i is the predicted score, λ is the regularization penalty weight, and θ is the vector of model parameters. The Huber Loss used here combines the advantages of mean square error and mean absolute error, so it is more robust to outliers and often used as the loss of regression tasks [70].

IV. EXPERIMENT

The experiments based on our collected dataset and MIT interview dataset [3] are conducted to analyze the effectiveness of the *Multi-modal Window Consistent Fusion (MWCF)* model. In the following sections, we will introduce the datasets and experimental settings at first. Then, we will show the results of performance comparison between *MWCF* and baseline models, as well as further ablation studies. Finally, a case study will be conducted to visualize the attention weights of different modalities to facilitate the understanding of effective components.

A. Datasets

The collected dataset: The asynchronous online interview platform developed for this project is shown in Fig. 7. We send the account of the interview system to the candidates through

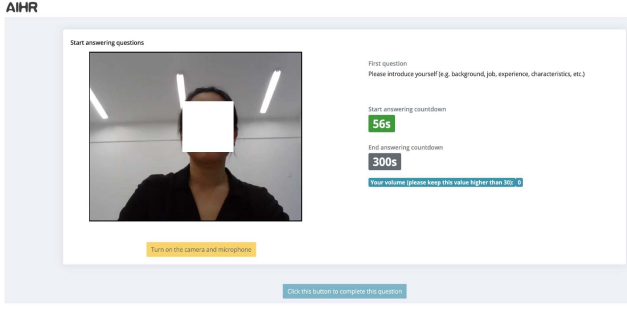


Fig. 7. Asynchronous online interview platform.

TABLE II
EXAMPLES OF THE KEYWORDS OF DIFFERENT SOFT SKILL ITEMS

Soft skill item	Keywords
Persuasiveness	Communication, Success, Requirements, Solution, Results, Objective, Challenge, Achieved, Difficulty, Innovate, Research, Beyond expectations, Leading demand, Potential demand, Insight into needs, Resource, Integration, Defuse, Negotiation, Clever, Reflect
Thinking Skills	Team, Project, Client, Market, Business, Plan, Overall, Formulation, Cost, Resources, Challenge, Risk, Target, Overall situation, Integration
Analytical Ability	Question, Market, Process, Objective, Data, Current, Background, Achieve, Analysis, Advance, Overall planning, Managing risk, Core, Key, Grasp the essence, Forecast, Survey
Conscientiousness	Project, Completed, Success, Task, Process, Time, Responsible for, Operation, System, Person in charge, Quality, Managing risk, Anticipating risk, Optimize resources, systematicness, Insist, Arrange work
Independence	Requirements, Overcome, Adjustment, Absorbed, Intervene, Communication, Trend, Channel, Formulation, Coordination, Division of labor, Self-care

SMS and email. The data collection has been introduced in Section III-A. After all the interviewees have finished their interviews, three HR consultants will give a comprehensive score to the candidates and grade the 18 soft skills for each question (except for self-introduction), which are divided into five grades: excellent, good, average, below average, and insufficient, with a score range of 0-5 points. We take the average score of the three HR consultants as the ground-truth label for each question. In addition, the HR consultants also select the most representative keywords under different soft skills as the prior knowledge base to participate in the training of the model. Table II gives some examples of the keywords in soft skills. Table III shows the details of the dataset. More detailed information on evaluation criteria and keywords for all soft skills is provided in the appendix, available online.

MIT interview dataset [3]: This is a video dataset containing 138 mock job interviews with students seeking internships. The average interview time is 4.7 minutes, and a total of 69 students are included. Each student received two interviews: one before the intervention of professional career counselors, and the other after the intervention of professional career counselors. Each

TABLE III
DETAILS OF THE DATASET

Attribute	Value
Number of valid interviewees	96
Number of soft skills	18
Number of <Answer Record/Soft skill>pair	3456
Average text length	912.7
Average number of words in text	519.4
Average number of words in text (removing stop words)	177.4
Average number of keywords in soft skills	41.7
Lowest score	0
Highest score	5

video in the dataset is scored by Amazon Mechanical Turk workers, and we average them to get the final score of the labels. The verbal content of the video is also manually transcribed by Turkers. We use the method mentioned in Section III-C above for feature extraction. Specifically, We extract verbal content and time stamps through the ASR system, and match the words manually transcribed by Turkers to reduce noise. When words are not matched, the time interval between the last successful match and the next successful match is used as the time stamp, and 300-dimensional glove [71] is used as the pre-trained word vector. Unlike the dataset collected in this paper, which is scored on the question level, the MIT dataset is scored on the interview level, so we use randomly initialized question vectors of (4) and keyword vectors of (5) in the topic generation module and jointly train with the model. Finally, we tested the model's performance on the RecommendHiring score of the interviewees.

B. Experimental Setting

In order to evaluate the prediction results, following the configurations of the research [20][42] [72], we use Pearson Correlation Coefficient (PCC) and Spearman rank Correlation Coefficient (SCC) as evaluation metrics. Considering the soft skill score data collected in the article was scored by the interviewer based on a five-point rating scale(excellent, good, average, below average, and insufficient), which has an ordinal relationship, we want to use the SCC to compare the monotonic relationship between the order list predicted by the model and the actual order evaluated by the HR consultant. Additionally, we use PCC to evaluate the linear relationship between model predictions and real results. To ensure that the indicators used are applicable to our research, we performed appropriate data pre-processing before conducting correlation analysis. When calculating SCC, we converted both the predicted data and the true labels into rank data to measure the monotonic relationship between the variables. The calculation method is as follows:

$$PCC = \frac{cov(Y, Y^{pre})}{\sigma(Y)\sigma(Y^{pre})} \quad (24)$$

$$SCC = \frac{cov(R(Y), R(Y^{pre}))}{\sigma(R(Y))\sigma(R(Y^{pre}))} \quad (25)$$

where Y is the ground-truth value of soft skills scoring and Y^{pre} is the predicted score. $cov()$ is the covariance of the variables, and $\sigma()$ indicates the standard deviation. $R(Y)$ is the rank converted by the variables Y .

In addition, we also use the Mean Absolute Error (MAE) and the Root Mean Square Error (RMSE) to measure the scoring error of the model

$$MAE = \frac{1}{N} \sum_{i=0}^{N-1} |y_i - y_i^{pre}| \quad (26)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} (y_i - y_i^{pre})^2} \quad (27)$$

where N is the number of samples, y_i denotes the ground truth for soft skills scoring, and y_i^{pre} is the predicted value of a model. The prediction accuracy for each fine-grained soft skill item is tested in the following experiments, and the final evaluation of the whole model is to average the metrics of all soft skill items.

In the experiment, we divided the participants into 10 parts, the training set, verification set and test set are divided according to the ratio of 8:1:1, so 20% of participants are used for validation and testing. Considering that different soft skills will have certain complementarities, we adopt a multi-task learning method in the training phase, and labels of different soft skills share the underlying parameters of the model. Specifically, we input keywords corresponding to the i^{th} soft skill in the topic generation module to predict the i^{th} soft skill scores. During training, the multi-layer perceptron parameters of (21) are independent, and other underlying parameters are shared. This effectively utilizes data from other tasks, thereby improving the performance of small sample tasks, and improving the generalization ability of the model in predicting soft skills.

The dimension of the pre-trained word embedding [59] is set as 300. For the words that do not exist in the vocabulary, their embedding vectors are initialized randomly in the range $[-0.01, 0.01]$. Due to the practical significance of the original scores of the dataset collected in this paper (respectively representing excellent, good, average, below average, and insufficient ratings), no normalization processing was performed. For the scores of the MIT interview dataset, we used the maximum and minimum normalization method to process them. Second, for the multimodal features extracted from both datasets, Z-score normalization was used to convert the independent variables into standard normal distribution in the experiment to improve the robustness of the model. The Adam optimizer [73] is used as the optimizer, and the learning rate is set to 0.001 with the batch-size of 128. In order to prevent over-fitting, the Xavier Uniform [74] is used to initialize the model parameters, together with the early stopping technique applied in the training process. Finally, the experiment uses the average effect of ten-fold cross validation as the result of the model.

C. Experimental Result

The following models are used as baselines to compare with our proposed *MWCF*:

- Random Forest(RF) [75]. RF is a classifier that uses multiple trees to train and predict samples. It was used in [76] to predict interviewees' openness, conscientiousness and

TABLE IV
INFLUENCE OF DIFFERENT MODALITIES ON THE MODEL IN THE COLLECTED DATASET

Methods	PCC	SCC	MAE	RMSE
RF-W	0.424	0.415	1.010	1.305
RF-A	0.405	0.412	1.119	1.317
RF-V	0.387	0.397	1.205	1.364
RF-WAV	0.461	0.463	0.995	1.298
LSTM-W	0.465	0.440	0.985	1.315
LSTM-A	0.457	0.447	0.994	1.289
LSTM-V	0.406	0.395	1.187	1.352
LSTM-WAV	0.504	0.498	0.946	1.283
HAN-W	0.458	0.430	0.977	1.305
HAN-A	0.445	0.422	0.994	1.311
HAN-V	0.401	0.394	1.056	1.324
HAN-WAV	0.519	0.522	0.932	1.250
MWCF-W	0.628	0.621	0.874	1.097
MWCF-A	0.588	0.578	0.898	1.115
MWCF-V	0.565	0.541	0.936	1.183
MWCF-WAV	0.671	0.665	0.852	1.088

TABLE V
INFLUENCE OF DIFFERENT MODALITIES ON THE MODEL IN THE MIT DATASET

Methods	PCC	MAE	RMSE
RF-W	0.542	0.188	0.220
RF-A	0.489	0.195	0.231
RF-V	0.297	0.244	0.285
RF-WAV	0.602	0.181	0.215
LSTM-W	0.551	0.184	0.226
LSTM-A	0.508	0.202	0.233
LSTM-V	0.357	0.229	0.267
LSTM-WAV	0.628	0.176	0.190
HAN-W	0.564	0.178	0.214
HAN-A	0.521	0.196	0.228
HAN-V	0.379	0.238	0.272
HAN-WAV	0.633	0.166	0.186
MWCF-W	0.577	0.183	0.215
MWCF-A	0.549	0.191	0.226
MWCF-V	0.394	0.237	0.276
MWCF-WAV	0.642	0.163	0.185

extroversion. In the following experiments, the average-pooling operation is applied on the input feature matrix S^x (x is A, V, or W) along the time dimension to achieve the multi-modal feature vectors, which are concatenated and fed into the RF model for final prediction.

- Long Short Term Memory(LSTM) [77]. The memory gate in the LSTM alleviates the problem of gradient vanishing and exploding in the sequence model, and enables the model to learn long-distance dependency in long time-series. In particular, [78] combined LSTM with the temporal pooling to predict the continuous emotion values.
- Hierarchical Attention Networks(HAN) [30]. HAN used a hierarchical attention structure to encode the features at both low and high levels respectively. On this basis, [18] improved the attention structure for the interview scene to predict the hirability of the candidates.

Furthermore, in order to explore the influence of different modalities, we denote the text modality, visual modality and acoustic modality model as W, V, and A respectively in the following experiments.

The results of the baseline models and our proposed *MWCF* model are shown in Tables IV and V. The suffix W, A, V

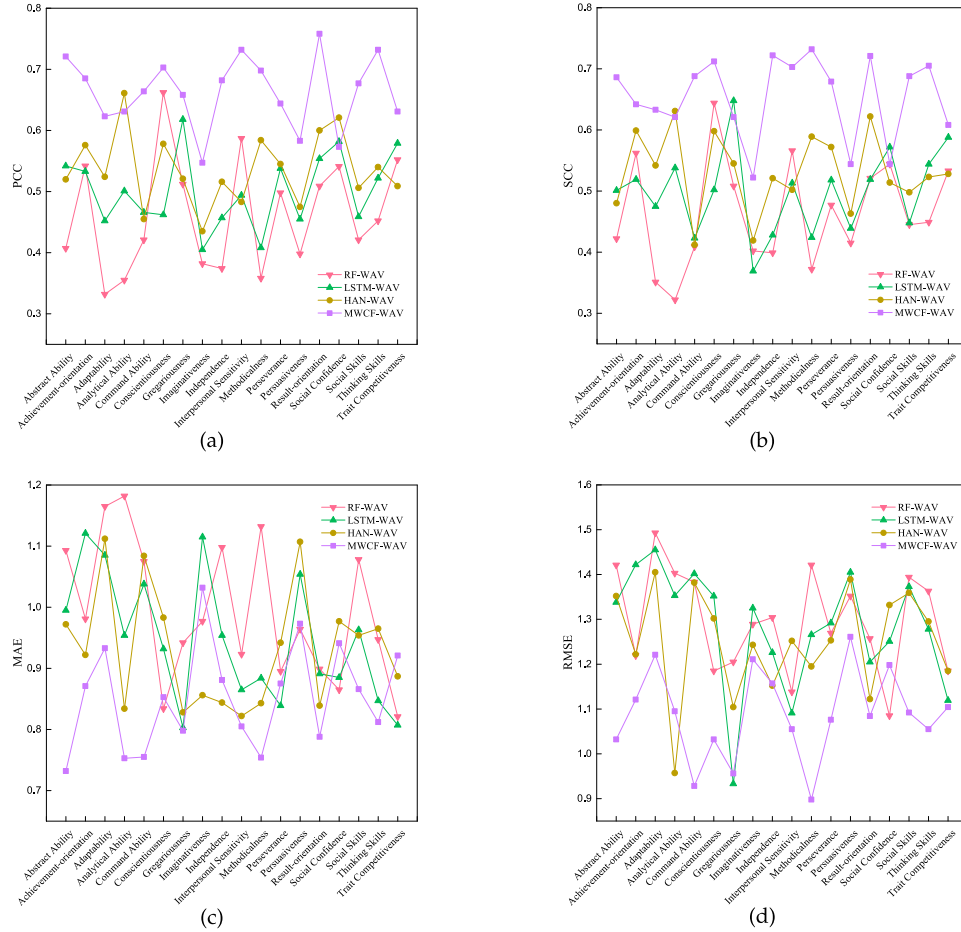


Fig. 8. Comparison of the results obtained in different soft skills using different models under multi-modal conditions: (a) Comparison of Pearson Correlation Coefficient (PCC), (b) Comparison of Spearman Correlation Coefficient (SCC), (c) Comparison of Mean Absolute Error (MAE), (d) Comparison of Root Mean Square Error (RMSE). The higher the values of sub-figure (a) and (b), the more accurate the model prediction is. The lower the values of sub-figure (c) and (d), the smaller the error of the model.

of different models represent different modalities. For example, LSTM-W means a uni-modal baseline using text modality only, LSTM-WAV means a multi-modal baseline using all three modalities. In order to achieve an overall evaluation of the prediction models in all test cases, we average the results of different soft skills for the collected dataset and show the experimental results in Table IV. It can be seen that for different models, the accuracy of multi-modal fusion model is significantly better than single-modal version. In particular, the PCC of multi-modal version of MWCF (MWCF-WAV) outperforms the text only model (MWCF-W), the auditory only model (MWCF-A) and visual model (MWCF-V) with 6.8%, 14.1% and 18.8% respectively in the collected dataset. Furthermore, different modalities have diverse effectiveness on soft-skill prediction. In particular, the text modality plays much more important role than other ones, as shown in the results of MWCF and all baseline models.

Tables IV and V also shows that the deep-learning based methods including MWCF, LSTM and HAN can achieve much better results than the traditional RF model under the multi-modal conditions. Compared with manual features based traditional machine learning methods, the deep hidden units of these deep-learning methods can better mine hidden features

for better understanding of multi-modal input. Furthermore, when equipped with the attention techniques, MWCF and HAN can achieve even better performance than the original LSTM model. Additionally, as observed in Table V, the performance gap between the HAN and MWCF models in the MIT dataset, both of which are hierarchical attention models, is not as substantial as the gap in the collected dataset. This is because we randomly initialize the keywords and question vectors in the topic generation module and learn from the data along with the model, which indirectly demonstrates the impact of introducing prior knowledge.

To further investigate the predictive performance of the MWCF model and the baseline model across different soft skills, we compared the effects of these models under multimodal conditions using the collected dataset, as depicted in Fig. 8. As shown in Fig. 8(a) and (b), the PCC and SCC score of our proposed MWCF model is much higher than the baseline models in most of the soft skills, which indicates that the prediction of MWCF is more consistent with HR consultants' scores. Fig. 8(c) and (d) further illustrate the MAE and RMSE errors of the models from another view, which show that the MWCF model has lower error than baseline models. In particular, although

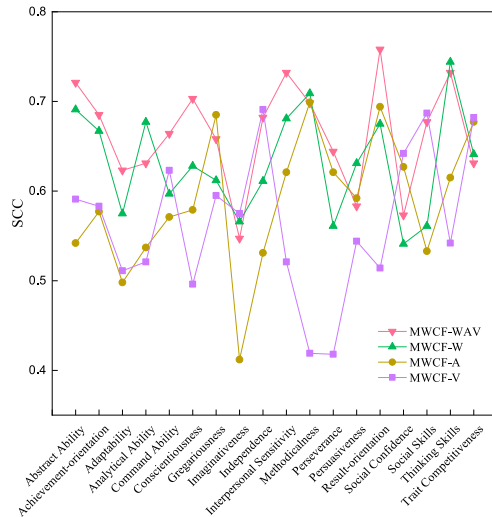


Fig. 9. Performance of *MWCF* models with different modalities under different soft skills.

both *HAN* and *MWCF* utilize multi-level attention methods to conduct information fusion, *MWCF* can achieve much better performance due to the effectiveness of the window-consistency attention mechanism, which integrates with the professional knowledge in the interview scene and aligns different modalities through window consistency.

To analyze the prediction effects of different modalities under different soft skills, we compare the performance differences between the multimodal and single-modal versions of *MWCF*. The results are shown in Fig. 9. The figure illustrates that various modalities exhibit diverse performances when it comes to different soft skills. Overall, multimodal models surpass unimodal models in predicting multiple soft skills, including command ability, achievement orientation, conscientiousness, etc. Multimodal models are able to capture an interviewee's capacity for delivering speeches and motivating team members, encompassing aspects such as oral expression, intonation, and rhythm. Additionally, the visual modality offers nonverbal communication information, such as facial expressions. Consequently, multimodal models extract valuable insights from multiple data sources, supplying richer contextual information. Text modality models excel at assessing thinking skills and analytical abilities since evaluating a person's discourse and logical reasoning can accurately reflect their cognitive prowess. In this context, textual information aids in analyzing language expressiveness and clarity, while audio and video may not contribute much supplementary information. Visual modality models demonstrate superior performance in the realm of social skills, as visual information helps models better comprehend individual behavior in varying environments. For instance, speaking with a smile may convey a friendlier and more sincere demeanor. Regarding persuasiveness, auditory modality models prove more effective, potentially because an interviewee's tone of voice, speaking pace, and expressiveness can better represent a person's emotions, professionalism, and attitude. These features significantly

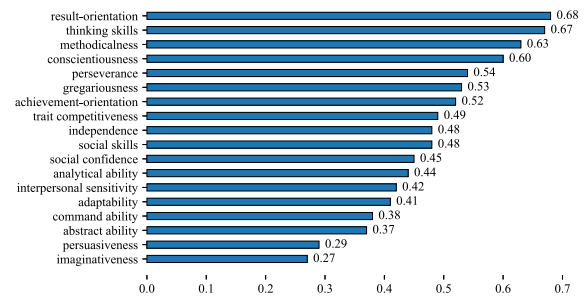


Fig. 10. Correlation between soft skills and overall scores.

influence persuasiveness. A candidate's confident and firm tone, accompanied by suitable pauses, may be more persuasive.

In addition, as can be seen from Fig. 9, due to the complex and diverse relationships in interpersonal interactions, the model has certain limitations in its evaluation of some soft skills. For example, being willing to persuade others and having rich imagination are more abstract abilities, which may be difficult to quantify or specifically assess, resulting in significant challenges for different modalities of models to fit them. Therefore, it is necessary to optimize the algorithm from multiple aspects, such as providing the model with more training data on these soft skills for different interviewees, and leveraging psychological research as well as context-based new word discovery to continuously refine and expand a more accurate fine-grained soft skill vocabulary.

One function of the automatic interview evaluation system is to assist interviewers in gauging, to a certain degree, an interviewee's performance in specific job-required abilities. This includes understanding their overall score and determining whether candidates are suitable for the position. Consequently, we analyzed the correlation between interviewers' overall assessment of interviewees and their soft skill scores. Fig. 10 illustrates a high degree of consistency between the interviewers' overall evaluation of candidates and their performance in result-orientation, thinking skills, methodicalness, and conscientiousness. It is essential to recognize that project outcomes are often a crucial component of job assessments. Results-oriented employees typically exhibit greater self-drive, often enhancing the team's overall performance, making them more appealing to interviewers. Simultaneously, conscientiousness, methodicalness, and conscientiousness help ensure the smooth progression of projects, reducing potential errors and omissions resulting from carelessness. They can promptly devise relevant plans, fostering an efficient and accountable work environment for the team. Generally, interviewers place more emphasis on candidates' problem-solving abilities and other team collaboration-related skills. Although persuasiveness and imaginativeness are identified in Fig. 10 as relatively less important in the overall evaluation by interviewers in this instance, they may prove beneficial in specific work scenarios, such as sales or creative industries. Thus, in actual interviews, it is often necessary to tailor the emphasis on soft skills based on the precise job requirements and the overall company culture. This ensures that the most

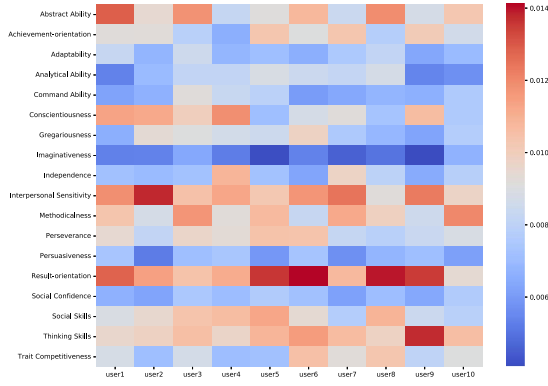


Fig. 11. Consistency weights for different soft skills of different users.

relevant skills, capable of generating value for the company, are accentuated.

To investigate which soft skills exhibit greater consistency across modalities, we randomly selected 10 users from the testset and created a heatmap that averaged the 30 time steps with the highest consistency weight for each user's different soft skills. As depicted in Fig. 11, the heatmap reveal that certain soft skills, such as result-orientation and interpersonal sensitivity, exhibit relatively high consistency weights between modalities. These skills may be more readily quantifiable, resulting in the better predictive performance of the model. Conversely, some soft skills display lower consistency weights, including imaginativeness and persuasiveness. These skills might manifest more abstractly in certain modalities, which can present challenges for the model when attempting to predict them. For such soft skills, it could be beneficial to explore more appropriate psychological features or provide additional training data related to these skills, enabling the model to learn more effectively and comprehensively.

D. Ablation Experiment

In this section, we study the effectiveness of different modules of *MWCF*, e.g. the topic generation module, and the window consistency fusion module. Specifically, the performance of following variation models are tested in the experiments:

- *MWCF- x w/o t* : This indicates the *MWCF* model without the topic generation module. A randomly initialized vector is used as c_i instead of the generated topic vector of (10). x here can be W, V, or A, which indicates the *MWCF* model with different modalities.
- *MWCF- x w/o k* : This indicates the *MWCF* model using the topic generation module with randomly initialized keyword list. Unlike the *MWCF- x* model, which uses a keyword list provided by the HR consultants (detailed soft skill keywords are included in the appendix, available online) and uses a pre-trained word vector to initialize the embedding matrix, the *MWCF- x w/o k* model randomly initializes the embedding matrix. This experiment is designed to verify the effect of prior keywords selected by

TABLE VI
PERFORMANCE OF VARIATION MODELS

Methods	x	PCC	SCC	MAE	RMSE
<i>MWCF-x w/o t</i>	W	0.483	0.471	0.963	1.277
	A	0.467	0.453	0.981	1.294
	V	0.432	0.439	1.014	1.314
	WAV	0.548	0.534	0.941	1.215
<i>MWCF-x w/o k</i>	W	0.524	0.516	0.933	1.231
	A	0.495	0.492	0.955	1.258
	V	0.474	0.482	0.977	1.265
	WAV	0.581	0.579	0.914	1.136
<i>MWCF-WAV w/o w</i>	-	0.644	0.638	0.884	1.107
<i>MWCF-WAV</i>	-	0.671	0.665	0.852	1.088

the HR consultants based on interview experience. x here can be W, V, or A.

- *MWCF-WAV w/o w* : This indicates the multi-modal *MWCF* model without using the window mask in the window consistency fusion module. *Window-Mask* in (14) is set as an all-ones matrix, and each element of which is equal to one. In this way, the window consistency attention degrades into the normal self-attention mechanism [29].

Table VI shows removing the topic generation module may cause a significant performance drop of the *MWCF* model, no matter in the single-modal or multi-modal cases. This confirms the effectiveness of the topic generation module, which enables the models to focus on the key information related to interview topics while ignoring most of the unimportant information in interviewees' long conversation.

Furthermore, the full version *MWCF* models has much higher performance than those without using the prior knowledge from HR consultants (*MWCF- x w/o k*). This shows the effectiveness of the keyword list preset by the HR consultants, which helps the machine to generate more proper prediction related to the interviewers' view points.

Finally, for the multi-modal version *MWCF-WAV*, by comparing the full version model with the variation model *MWCF-WAV w/o w* , we can observe the significant performance drop by removing the window mask. This shows the effectiveness of the window-consistency attention compared with traditional global self-attention mechanism [29].

E. Sensitivity Analysis of Window Size

To analyze the effectiveness of using different window sizes, we conduct experiments on the *MWCF* models with the window size from 5 to 200, and show the results in Fig. 12. Specifically, when the window size is 200, it is equivalent to a model without window restrictions. It can be observed from Fig. 12 that when the window size is small, the performance of the model is even worst than the model without window restrictions. This is because the multi-modal expressions are not always synchronized strictly, e.g. when an interviewee introduces an interesting thing, he may smile after telling the story. When the window size is too small, the consistency of multi-modal expressions may be hard to observe while deploying the window based topic attention on each modality. In addition, we can observe that the model works best when the window size is about 30, and the performance exceeds the model without window limitation.

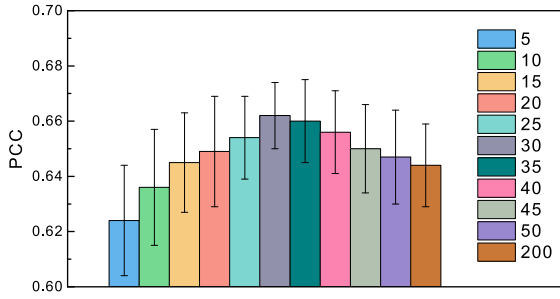


Fig. 12. Influence of window size on MWCF model.

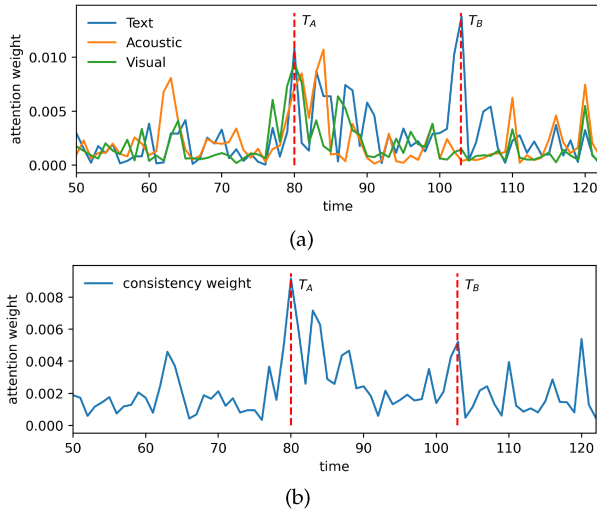


Fig. 13. Visualization of attention in multi-modal windows.

Actually in the interview, the average speed of speech is about 250 Chinese Characters per minutes, which contains about 140 Chinese words. That means 30 words are corresponding to about 13 seconds. The experimental results indicate that the speech content within the distance of 13 seconds is quite correlated and usually express a close topic about the interview. Thus applying the local attention on that scope can efficiently capture the correlation to strengthen the description about the interviewee. This confirms that the window with appropriate size can effectively limit the attention range of the model and reduce the interference of noisy information.

F. Case Studies

In this section, we will introduce some typical case studies to better explore the effectiveness of multi-modal window consistency. In particular, Fig. 13 visualizes an interviewee's multi-modal attention of the soft skill prediction of conscientiousness when the interviewee is answering the question "what do you think is the most successful thing you have done". Fig. 13(a) shows the attention on each modality, and Fig. 13(b) shows the weighted multi-modal consistency attention. We focus on two moments indicated as the time T_A and T_B in Fig. 13. It shows that the attention of different modalities at the time T_A is more consistent than T_B , which causes a higher consistency

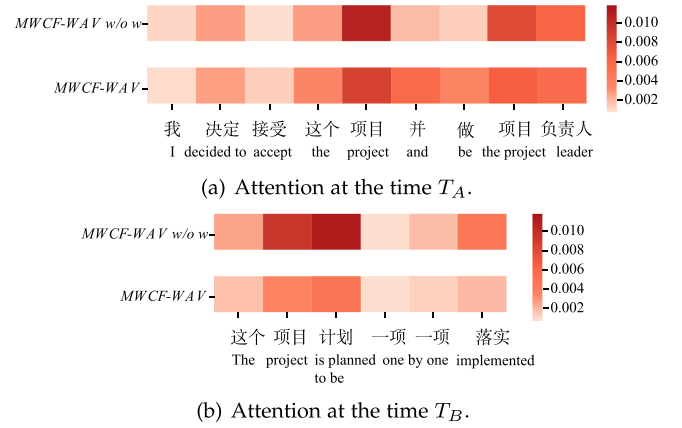


Fig. 14. Attention of the text modality. The HR Consultant scored this employee as 3.5 points, the model without window mask (*MWCF-WAV w/o w*) scored 2.5 points, and the full model (*MWCF-WAV*) scored 3.2 points.

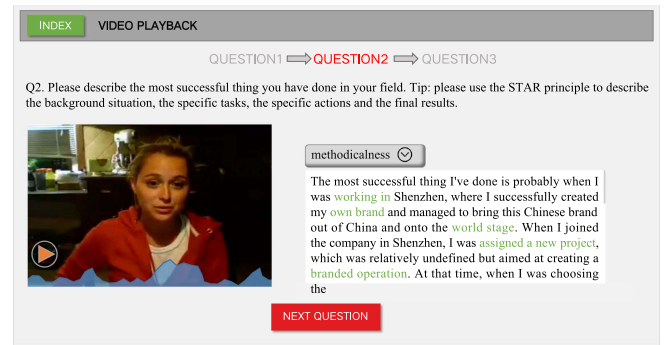


Fig. 15. Interpretability of models in interview systems.

weight at the time T_A as shown in Fig. 13(b). That means T_A is a moment when the employee's face expression and tone are more consistent with what he says, and is more possible to reveal his personal character veritably. For example, when an interviewee describes a successful event he or she is very proud of, but there is no change in expression or tone of voice, this may indicate that the interviewee has little or no involvement in the matter.

As shown in Fig. 14, to observe the attention on the employee's speech content more intuitively, we plot the attention weight of each word said by the employee at the time T_A and T_B . Two sentences at the time T_A and T_B are analyzed here. Without using the window-consistency attention mechanism (*MWCF-WAV w/o w*), the words in both cases (Fig. 13(a) and (b)) are assigned with high attention. By using the full model *MWCF-WAV*, the case at the time T_A is assigned with higher attention, which is consistent with Fig. 13(b). Meanwhile Fig. 14 also shows that "project", "leader", "plan" and "implement" are assigned with higher attention than other words, which have similar semantics with "project", "responsible for" and "arrange work" in the keyword list of Conscientiousness given by the HR consultants in Table II.

As shown in Fig. 15, we enhance the interpretability of the model within the system by combining consistency attention weights with video clips. Specifically, we highlight words with

higher attention weights in the model's predictions alongside the corresponding video segments, as seen in the higher position of the wavy line at the bottom of the video and green words in the system, which represent high weights. Furthermore, when the interviewer selects different soft skills, the highlighted parts will also be adjusted accordingly. This approach helps users better understand the prediction process of the model and why certain words and segments are considered more important. However, it is important to note that although higher weights typically indicate that a word or segment has a greater influence on the prediction process, they should be treated with caution in real-world applications. Model weights are only a reference and may not necessarily be the real reason of the model's predictions. Therefore, when analyzing and evaluating the predicted results, the interviewer should consider other factors and contextual information to obtain more accurate and reliable conclusions. In addition, the interviewer can provide feedback on samples through the system periodically, which may help to some extent in mitigating potential biases within the model.

V. CONCLUSION

In this article, we propose a novel *Multi-modal Window-Consistency Fusion* network, namely *MWCF*, to capture the expression consistency of different modalities in short-time windows and re-weight the language signals to enhance important portions in verbal clues. Meanwhile, in order to enhance the interpretability of the evaluation model, we propose an attention based topic generation model to integrate with the keywords based professional knowledge collected from HR consultants. Furthermore, to verify the effectiveness of the model, we have built an online asynchronous interview platform to collect real-world interview dataset and invite three professional HR consultants to score the candidates' soft skills. Experiments show that our method can achieve superior performance compared with the baseline models and provide an important reference for interviewers to make decisions.

For the future work, we will introduce some multi-task learning methods to make the model learn more abundant external knowledge. In addition, we will study how to make better use of unlabeled data for better unsupervised domain adaptation.

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